

Multiple Choice

1. **Answer:** D

Options: A) Melting point; B) Boiling point; C) Polarizability; D) First ionization energy

Note: The first ionization energy of argon is lower than that of neon because argon has a larger atomic radius and more electron shielding, which reduces the effective nuclear charge experienced by its outermost electrons, making them easier to remove.

2. **Answer:** D

Options: A) 10.8×10^{-5} ; B) 4.11×10^{-5} ; C) 3.43×10^{-5} ; D) 1.71×10^{-5}

Note: The correct answer is 1.71×10^{-5} for the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field.

2. **Answer:** D

Options: A) $Q = 1012$; B) $Q = 2012$; C) $Q = 3012$; D) $Q = 6012$

Note: The Q-factor, defined as the ratio of the resonant frequency to the bandwidth, indicates the energy loss relative to the stored energy in the cavity, and for an X-band EPR cavity with a bandwidth of 1.58 MHz, a Q-factor of 6012 suggests a high level of resonance and low energy loss.

3. **Answer:** A

Options: A) 3; B) 5; C) 8; D) 4

Note: The correct answer is 3 because the ^{55}Mn nuclide has a nuclear spin of $5/2$, which corresponds to three allowed energy levels based on the possible projections of the nuclear spin in a magnetic field.

4. **Answer:** B

Options: A) The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.; B) The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.; C) The $\text{MnO}_4^- : \text{Mn}_2^+$ ratio is 3:1.; D) The $\text{H}^+ : \text{I}^-$ ratio is 2:1.

Note: The correct answer is based on the stoichiometry of the balanced redox reaction, where 6 electrons are transferred from 5 iodide ions (I^-) to one permanganate ion (MnO_4^-), resulting in a ratio of 6:5 for MnO_4^- to I^- .

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because, despite potassium (K) being a larger atom than scandium (Sc) in the periodic table, the ionic radius of K^+ is larger than that of Sc_3^+ due to the greater effective nuclear charge and loss of three electrons in Sc_3^+ , which

results in a stronger attraction of the remaining electrons towards the nucleus, leading to a smaller ionic radius.

7. **Answer:** False

Options: A) True; B) False

Note: Infrared spectroscopy is not a light-scattering technique; it involves the absorption of infrared light to analyze molecular vibrations.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because NH_3 (ammonia) is actually a weak base in liquid ammonia, and it does not have a high tendency to donate protons; rather, it tends to accept protons, making it a Lewis base rather than a strong proton donor.

9. **Answer:** True

Options: A) True; B) False

Note: Silicon primarily bonds with oxygen in the Earth's crust, forming silicon dioxide (silica) and various silicate minerals, which account for the majority of silicon found in nature.

10. **Answer:** False

Options: A) True; B) False

Note: Doubling the volume of a buffer solution by adding water dilutes the concentrations of the buffering agents but does not significantly change the pH because buffers are designed to resist pH changes within a certain range.

Long Form Response

11. **Answer:** The differing relaxation times, T_1 and T_2 , are influenced by various factors, particularly the molecular motions occurring at the Larmor frequency. T_1 , or longitudinal relaxation time, is primarily associated with the transfer of energy from the nucleus to its surrounding lattice, which is less sensitive to rapid molecular motions. In contrast, T_2 , or transverse relaxation time, is significantly affected by these molecular motions because it involves the dephasing of nuclear spins due to interactions with other spins and fluctuating magnetic fields at the Larmor frequency. As molecular motions increase, they can disrupt the coherence of the nuclear spins more rapidly, leading to a shorter T_2 time. Thus, while T_1 reflects energy exchange processes, T_2 highlights the impact of molecular dynamics on spin coherence.

Note: The answer correctly identifies that T_1 relaxation is primarily influenced by energy transfer to the surrounding lattice, which is less affected by rapid molecular motions, while T_2 relaxation is more sensitive to these motions due to the dephasing of nuclear spins at the Larmor frequency.

12. **Answer:** Ionic strength is a measure of the total concentration of ions in a solution, which influences various chemical properties, including solubility and reaction rates.

A 0.050 M solution of AlCl_3 exhibits higher ionic strength compared to other ionic solutions due to its complete dissociation into three ions: one aluminum ion (Al^{3+}) and three chloride ions (Cl^-). This results in a total effective concentration of ions of 0.050 M AlCl_3 contributing $0.150\text{ M} = 0.200\text{ M}$ ionic strength. In contrast, a 0.050 M solution of a salt like NaCl only contributes 0.100 M ionic strength. Note: correct because it accurately calculates the ionic strength of a 0.050 M solution of AlCl_3 by considering

End of Answer Key